Expression After SIMB

The Knockout-Expression Strand: What the Campaign Established,

Read Round by Round

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Abstract

One strand of the SIMB retrospective, knockout expression on Kemmeren and Sameith, became the campaign that followed it: four rounds on IGB and Delta between 2026-08-31 and 2026-09-08, all under the masked-label objective of project `torchcell_019_expr_v9`. Every score is one statistic, the maximum of a centered 5-epoch rolling mean of the per-gene validation Pearson, read from the full history of every run and placed against eight identical-config replicates of the incumbent, whose spread at the full budget is 0.0222 around a mean of 0.1965.

What is established. The distributional head is not load-bearing: a point head collapses, the CRPS head collapses two times in three, and the Laplace-CRPS head ties the quantile head to 0.0005. The validation loss bottoms out at a few hundred epochs in every run while the metric keeps rising to beyond 19,000, so the long budget buys Pearson on a rising objective. Training on the metric itself does not escape that: pure Pearson at batch 32 collapses to constant outputs at every seed, and the batch-64 variant that survives sits on the incumbent band rather than above it. The one factor that moves the score by more than the replicate spread is the content of the gene embedding, ProtT5 over random by 0.07 at 1,000 epochs; trunk depth, readout width and weight decay do not. The per-gene readout is +0.03 over its in-round reference at two seeds, under the resolution of that round, and the pair term alone is a null. Four of four packed five-day tasks died of host memory.

What is not. Nothing measured beats the incumbent's replicate mean at any budget; the incumbent's own embedding, `calm`, has not been placed against ProtT5; and no design run so far resolves a gap under about 0.04 at the full budget.

How to read this. section 1 is the state of the strand: every claim with its script, its statistic and its sample size, in the order the evidence arrived. section 2 is the readout of the three rounds that returned on 2026-09-08. section 3 is what the evidence leaves open. The design and launch of the rounds, and the retrospective they grew out of, are in `notes-tex/019-simb-multimodal/` (sections 9 and 10 there), which is not repeated here.

Working note: `notes/experiments.019-simb-multimodal.expression-strand-retrospective.md`

Generating scripts: `experiments/019-simb-multimodal/scripts/`

Leaderboard: `experiments/019-simb-multimodal/results/round_leaderboards.csv`

Section status: ✓ ready ■ author-reviewed, keep stable ✗ not done, agents may edit freely

Contents

1 The state of the strand, claim by claim ✗

The task is the one the SIMB retrospective defined: predict the log-ratio expression of every measured gene in a deletion strain from the deletion, on the Kemmeren and Sameith compendia under a fixed partition, scored by the per-gene Pearson across held-out strains and averaged over genes. The replicate ceiling for that score is 0.775. Everything below is read from the leaderboard puller (`pull_round_leaderboards.py`) over project `torchcell_019_expr_v9`, which is the masked-label objective, or from a script named in the paragraph. The statistic throughout is `roll_max`: the maximum of a centered 5-epoch rolling mean of `val/expression/pearson_per_feature`, an upward-biased order statistic whose bias grows with the epochs run, and which cancels in a contrast between runs scored the same way.

1.1 The incumbent and its spread ✗

The incumbent is the quantile head under the pinball loss, learning rate 3×10^{-4}, dropout 0.1, six transformer layers at hidden width 90, the nine-graph attention mask, `calm` gene embeddings, batch 32, 9,900 epochs. Eight identical-config runs of it score $\mathbf{0.1965 \pm 0.0222}$; the three highest leaderboard entries, 0.238 to 0.243, are one resumed lineage of that configuration continued to 19,000 epochs and are the maximum of those eight draws, not a different model. Baselines that need no GPU (`expression_baselines.py`): per-gene mean 0.0000, bilinear on ProtT5 0.1040, embedding-neighbor 0.0908. The incumbent clears the bilinear by 0.093, about 4.2 standard deviations. On squared error no live run beats the per-gene mean at its Pearson peak: the best `nmse` at peak is 1.005.

1.2 The head is not load-bearing ✗

The objective round (IGB, launched 2026-08-31, jobs 2368333, 2368337, 2368339; three replicates per challenger at the incumbent's budget) asked whether the quantile head does anything a point head or another proper scoring head would not.

head	n	collapsed	live mean	live sd	live max	min `nmse` at peak
`point`	6	6	–	–	–	–
`crps`	6	4	0.1697	0.0094	0.1764	1.075
`laplace_crps`	6	0	0.1947	0.0167	0.2154	1.025
`quantile`	20	0	0.1952	0.0258	0.2430	1.005

Table 1: The objective round. Collapse is a final Pearson numerically zero: the network emits one constant per gene. The point head collapses every time, CRPS two times in three, the Laplace-CRPS and quantile heads never. Between the two that survive the gap is 0.0005 against a detectable gap of about 0.042 at three replicates versus eight.

What the head buys is not collapsing. Between the two heads that do not collapse the choice was never load-bearing, and the round was sized so that only a large effect could have shown; the honest statement is that the heads are indistinguishable at any budget that can be afforded.

1.3 Replicate spread is a function of budget, and not a monotone one ✗

The same eight identical-config runs, rescored on the prefix of their own curves (`short_budget_spread.py`):

Two traps were hit measuring this and are recorded so the next reader does not hit them: a resumed run passes an epoch filter and inherits its parent's config, so it masquerades as a replicate and is identified only by where its curve starts; and a whole-curve-maximum collapse test reads a run that collapses late as healthy, so collapse is tested on the last value.

1.4 A short screen does not rank like a long run, within one configuration ✗

Spearman between the 700-epoch and 3,000-epoch `roll_max` of the same run (`budget_rank_preservation.py`): -0.139, $p = 0.53$, $n = 23$; top-3 overlap one of three, top-5 one of five. After resume exclusion the corpus is essentially one configuration, so this measures within-config persistence only: a run's early rank among its own replicates says nothing about its late rank. Whether a short screen orders different configurations correctly is answerable only on a multi-config corpus, and the v10 grid (section 2.1) is the first one.

epochs	mean	sd	main effect, 32 v 32	cell, 16 v 16	arm, 8 v 8
350	0.1198	0.0096	0.0067	0.0095	0.0135
500	0.1223	0.0058			
700	0.1411	0.0166			
1,000	0.1609	0.0099			
1,500	0.1670	0.0117			
2,000	0.1745	0.0151			
2,800	0.1821	0.0175			
4,000	0.1883	0.0171			
6,000	0.1917	0.0177			
9,900	0.1965	0.0222			

Table 2: Mean and spread of the incumbent's eight replicates at truncated budgets, from 500-sample curves. The spread is not monotone in budget: 700 epochs is the noisiest point below 2,000. The last three columns are the detectable gap at the 350-epoch spread for the three contrast sizes a 2^4 grid affords, the same arithmetic that sized the v10 grid at 1,400 epochs and two seeds rather than 700 and four.

1.5 The validation loss bottoms out early and the metric keeps rising ✗

Full per-epoch history of every live v9 run with a proper-scoring head, 23 runs of 1,000 epochs or more that neither collapsed nor resumed, read 2026-09-08 (`loss_min_vs_pearson_peak.py`, fig. 1). The median epoch of the `val/loss` minimum is 481 (range 203 to 3,097) and no run bottoms out by epoch 100. The median final loss sits 11.5 % above its minimum. Every run's Pearson peak (median epoch 2,488) comes after its loss minimum, at a loss already 7.1 % above the floor, and the median Pearson gained between the two is +0.054. The long budget buys Pearson on a rising validation objective, which is the fact the metric-aligned round was built to confront. The eight Pearson-objective runs are excluded here by construction, since their loss and metric coincide on train.

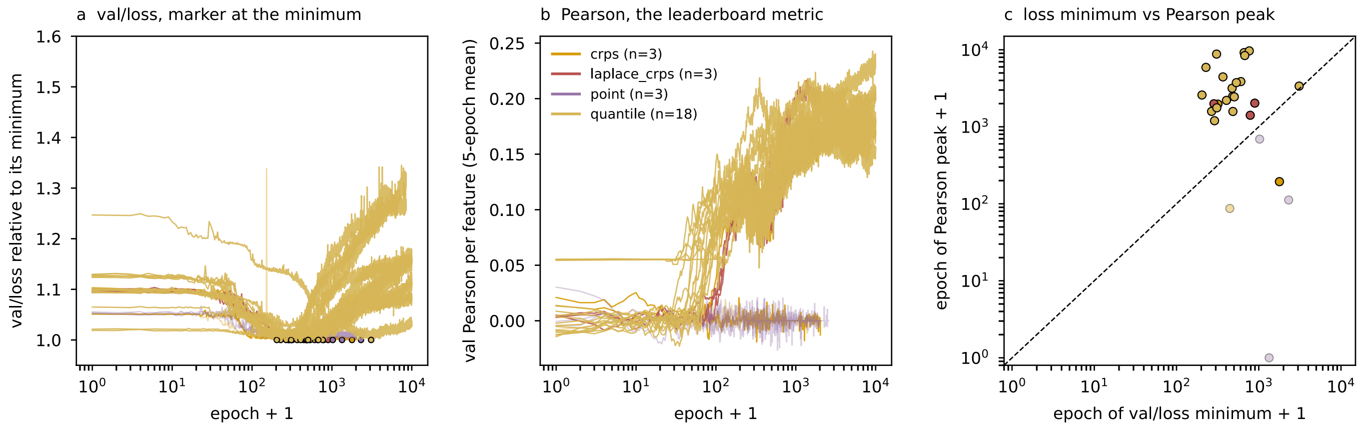

Figure 1: Where the loss bottoms out and where the Pearson peaks. **a**, validation loss relative to its minimum, one curve per run, colored by head. **b**, the 5-epoch rolling mean of validation Pearson. **c**, the epoch of the loss minimum against the epoch of the Pearson peak; every point lies above the diagonal. Generated by `loss_min_vs_pearson_peak.py`.

1.6 Training on the metric does not resolve it ✗

Two heads were added to `torchcell/losses/distributional.py` on 2026-09-06: `pearson`, whose loss is one minus the mean per-feature Pearson over the genes hidden at the current unmasking step, each gene's correlation taken over the strains in the batch where it is hidden, with columns of fewer than three scored rows or constant prediction or target dropped as the metric drops them; and `pearson_mse`, the same plus the masked squared error at unit weight as a scale anchor. Both are point-shaped, carry no PIT and no calibration keys, and the pure loss is scale-free, so its `mse`, `nmse` and spread ratio are not comparable to any other arm's. Eighty-six unit tests and a container canary (2375312) preceded the launch. The readout at day two is section 2.3: collapse at every batch-32 seed of the pure loss, at two of three seeds of the anchored one, and parity with the incumbent for the batch-64 survivors.

2 The readouts, 2026-09-08: three rounds, one large effect ✗

Three rounds returned within two days of each other: the 2^4 factorial on Delta, the mechanism round on IGB (both designed in the SIMB retrospective's launch section, `notes-tex/019-simb-multimodal/sections/10-launch.tex`), and the first two days of the metric-aligned objective round, which trains on $1 - r$ directly (section 1.6). Every score below is `roll_max`, the maximum of a centered 5-epoch rolling mean of `val/expression/pearson_per_feature`, read from the full per-epoch history, and every contrast is at a MATCHED budget computed as the smallest final epoch across the runs being compared. The scoring rule is an upward-biased order statistic; it is the same rule on every side of every contrast, so the bias cancels in a difference and not in an absolute number.

2.1 The factorial: embedding content is the only factor that moves the score ✗

Job 21830323 ran 16 cells at two seeds, one run per GPU, four per node. Five of the eight array tasks completed 1,400 epochs; three hit the two-day wall between epochs 990 and 1,198, so the matched budget is epochs ≤ 990.

factor	contrast (level 1 − level 0)	effect	t	effect, 31 runs	t
embedding	`prot_T5_all` − `random_1024`	+0.068	+7.8	+0.076	+17.3
trunk	$L{=}2, h{=}45$ − $L{=}6, h{=}90$	−0.012	−1.4	−0.018	−4.0
readout	linear − two-layer	+0.007	+0.8	+0.002	+0.5
weight decay	10^{-4} − 10^{-8}	+0.005	+0.5	−0.000	−0.1
embedding × trunk		+0.017	+2.0	+0.013	+2.9

Table 3: Main effects of the v10 grid at epochs ≤ 990, 16 runs a side. The error is the pooled within-cell replicate standard deviation, 0.0246 on 16 degrees of freedom over all 32 runs (standard error per effect 0.0087), and 0.0122 on 15 degrees of freedom once the one run that never left the chance band is dropped (standard error 0.0044). The other five two-way interactions have $|t| < 1.3$ in both readings.

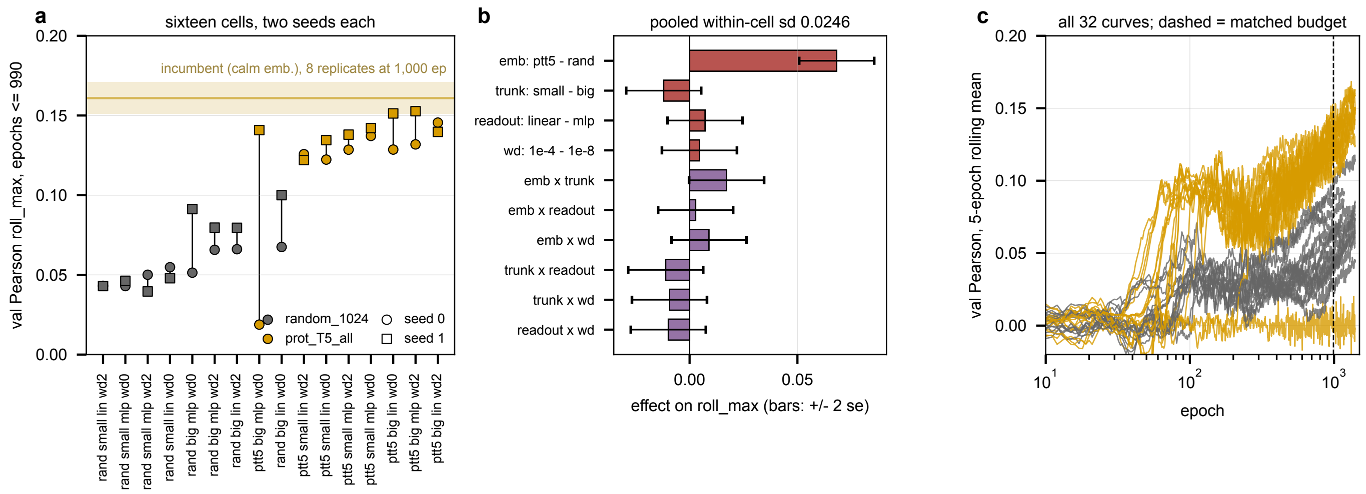

Figure 2: The v10 grid at a matched budget. a, every cell's two seeds at epochs ≤ 990, sorted by cell mean and colored by embedding, against the eight identical-config incumbent runs at 1,000 epochs (0.161 ± 0.010, `calm` embeddings, band is one standard deviation). **b**, main effects and two-way interactions with ± 2 standard errors from the pooled within-cell spread. **c**, all 32 validation curves; the dashed line is the matched budget. Generated by `v10_grid_factorial.py`.

Embedding content is worth 0.07, three times the replicate spread.. The ProtT5 cells average 0.129 against 0.061 for random vectors of the same width, and no cell of one level overlaps any cell of the other (fig. 2a). The contrast was built as a content test at matched width 1,024, so it says the content matters and says nothing about `calm`, the incumbent's embedding, which is not a level of the grid. The one healthy draw of the incumbent-with-ProtT5 cell scores 0.141 against the incumbent's 0.161 ± 0.010 at the same budget, one draw against eight, and that is not a resolved gap in either direction.

Readout width and weight decay are measured nulls.. Both effects are under 0.008 against a resolution of about 0.02 at two standard errors. The trunk is a small effect in favor of the deeper one, −0.012 over all runs and −0.018 once the stuck run is dropped, and it is sharper under ProtT5 (the interaction). Of the four factors chosen to act on the train-validation gap, one acts.

One run in 32 never learned.. Cell 1 seed 0 (`te8272kk`: ProtT5, L=6, two-layer readout, weight decay 10^{-8}) sits at 0.019 at its best and −0.007 at epoch 1,399 with `nmse` 1.010, while its seed-1 twin is the grid's best single run at 0.169. That pair alone carries most of the pooled variance, which is why both readings are given. Its cause is not measured.

2.2 The mechanism round: per-gene readout +0.03, under the round's resolution ✗

All four array tasks of the round ended in the cgroup out-of-memory handler at host RSS 61 GB against a 60 GB request, after 3.5 to 4.6 days. In each task one of the two packed runs was killed and the other finished its 8,500 epochs, so the matched budget is set by the shortest survivor, epochs $\leq$ 4,079. The two seed-1 runs abandoned on cabbi at epoch $\approx$1,595 when that half moved to the A40 node are superseded, not replicates, and are dropped.

arm	seed 0	seed 1	paired difference vs `R_ref` (s0 / s1)	mean
`R_ref`	0.188	0.169		
`R_basis64`	0.175	0.200	−0.014 / +0.030	+0.008
`R_pergene`	0.189	0.224	+0.001 / +0.054	+0.028
`R_pergene_basis64`	0.214	0.199	+0.026 / +0.029	+0.027

Table 4: The mechanism round at epochs $\leq$ 4,079, paired within seed against the in-round reference. Seed 0 ran on cabbi Ada cards, seed 1 on one A40 node, so GPU type is a seed-level block. The incumbent's eight replicates score 0.188 ± 0.017 at 4,000 epochs and both `R_ref` draws fall inside that band.

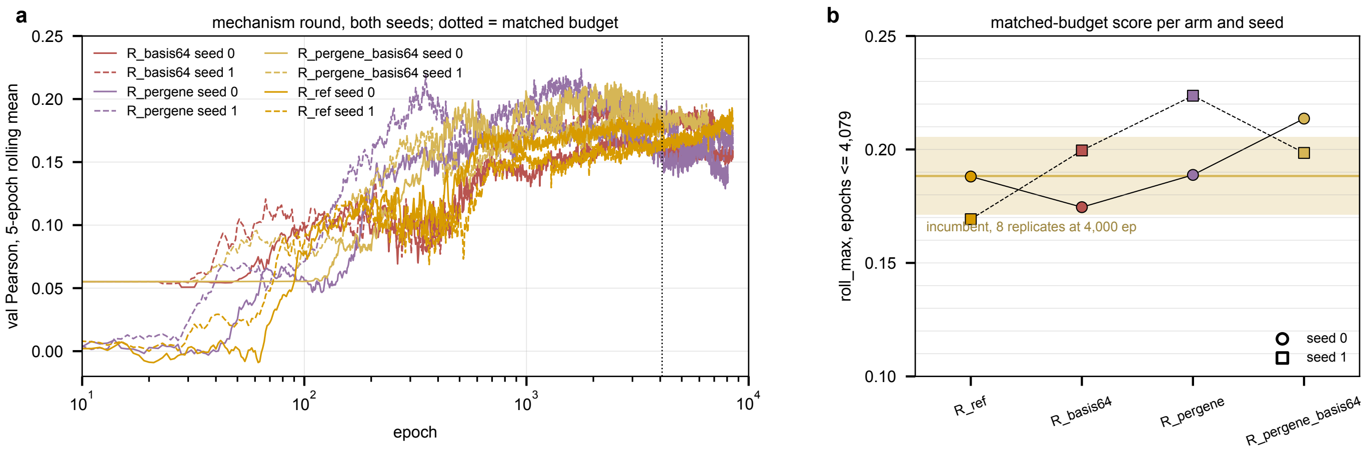

Figure 3: The mechanism round. **a**, validation Pearson of all eight runs (solid seed 0, dashed seed 1); the dotted line is the matched budget. **b**, the matched-budget score per arm and seed against the incumbent band at 4,000 epochs. Generated by `mech_round_readout.py`.

The pair term alone is a null.. Its two paired differences disagree in sign and average +0.008. The per-gene readout arms average +0.027, and the combined arm is positive at both seeds, but the round was sized to resolve about 0.06 at two pairs (table 2), so this is a direction and not a result. What every curve does show is the dynamics: the per-gene arms reach their peak between epochs 1,200 and 2,600 and give part of it back (`R_pergene` ends at 0.137 and 0.170 against peaks of 0.189 and 0.224), while the reference and basis arms are flat or still rising at 8,500. Whether the give-back is overfitting of the extra per-gene parameters is a hypothesis and is not measured.

A tagging defect, found in the readout.. The arm script carried a wave-4b `R_ref` branch ahead of the wave-5 one; a shell `case` takes its first match, so every reference run of this round is tagged `stage-wave4b`. The readout selects by arm and config tag, and the shadowing branch is removed.

2.3 The metric-aligned round at day two: collapse is the rule at batch 32 ✗

Three arms train on $1 - r$ directly, the mean per-feature Pearson over the genes hidden at the current unmasking step (`pearson`), the same plus a masked squared error at unit weight (`pearson_mse`), and the pure loss at batch 64 (`pearson_b64`), on IGB cabbi from 2026-09-06. The two batch-32 arms run three seeds at two runs per card, the batch-64 arm two seeds solo. They were read at 1.9 of their 5 days, so every number here is partial and the readout script records the epoch reached.

arm	seed	epoch	`roll_max` @ epoch	last	on the floor from	spread ratio at end
`pearson`	0	3,902	0.145 @ 215	0.000	584	0.000
`pearson`	1	3,886	0.133 @ 151	0.000	2,203	0.000
`pearson`	2	3,021	0.137 @ 242	0.000	680	0.000
`pearson_mse`	0	3,899	0.137 @ 138	0.000	360	–
`pearson_mse`	1	3,888	0.194 @ 3,193	0.168	alive	0.43
`pearson_mse`	2	3,023	0.035 @ 5	0.000	14	–
`pearson_b64`	0	6,051	0.194 @ 1,926	0.163	alive	0.80
`pearson_b64`	1	6,049	0.186 @ 2,419	0.138	alive	0.73

Table 5: The metric-aligned round, partial. "On the floor from" is the first epoch after which the 5-epoch rolling mean of the validation Pearson stays under 0.02; the seed-1 `pearson` run is near zero from epoch ≈250 and flickers above the line until 2,203. The spread ratio is predicted over true standard deviation (`pred_sd_ratio`). The incumbent's eight replicates score 0.188 ± 0.017 at 4,000 epochs and 0.192 ± 0.018 at 6,000.

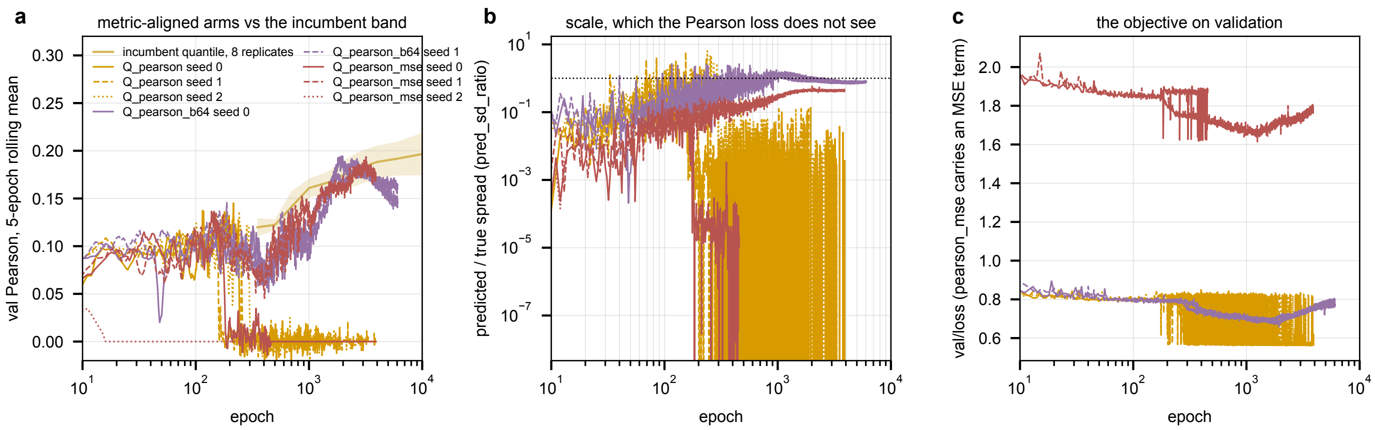

Figure 4: The metric-aligned round at day two. a, validation Pearson of all eight runs against the incumbent replicate band. **b**, predicted over true spread, which the Pearson loss cannot see; a run whose outputs go constant falls off the bottom. **c**, the objective on validation; `pearson_mse` carries the squared-error term and sits on its own scale. Generated by `pearson_round_readout.py`.

Five of six batch-32 runs are on the floor, and the loss does not see it.. The three pure-Pearson runs peak between 0.13 and 0.145 at epochs 150 to 240 and then fall, their predicted spread dropping from about 0.3 to 10^{-7} over the same epochs (fig. 4b): the network's output goes constant per gene. Their validation loss holds near 0.56 throughout, because the loss drops a constant column as invalid and averages over the columns still varying, so it reads $1 - 0.44$ on a handful of genes while the metric reads 0 on all of them. The failure mode the design named is the one that occurred, with the difference that the loss never reads its connected zero. The squared-error anchor at unit weight did not prevent it: `pearson_mse` collapsed by epoch 14 at one seed and 360 at another, and its surviving seed is alive at 0.194 and rising, inside the incumbent band.

The batch-64 runs survive and sit on the band, not above it.. Both are at 6,050 epochs with spread ratios of 0.73 and 0.80, `roll_max` 0.194 and 0.186 against the incumbent's 0.192 ± 0.018 at that budget, peaked near epochs 1,900 to 2,400 and drifting down since. Batch size and solo placement are confounded in that arm, so why it survives is not measured; the in-batch correlation being taken over twice as many strains is a hypothesis. Nothing in the round beats the quantile head at any budget read so far, and the question it was launched to answer, whether the metric's late rise is generalization gap or objective disagreement, is answerable only on the live runs after the wall.

2.4 Packed five-day runs die of host memory ✗

Four of four packed tasks that ran past three and a half days ended `OUT_OF_MEMORY` (`2369697` 0–1, `2371531` 2–3), at `MaxRSS` 61.4 and 61.1 GB. The Pearson-round packed tasks read 21.3 GB at 1.9 days and the solo batch-64 tasks 17.2 GB. Hypothesis (untested): resident memory grows roughly linearly through a run, in which case the packed Pearson tasks reach the 60 GB line near day five, at the edge of their wall. What grows is not identified; the `file_system` sharing strategy introduced in `f1bfa95b` to stop the file-descriptor exhaustion is the obvious suspect and is unverified. The memory of a running job cannot be raised, and the two rounds that finished lost one run per task to it rather than the whole task.

3 What the evidence leaves open ✗

Each item is a decision the measurements above do not make, stated with the measurement that would.

Which number is the target.. Stopping on the validation loss says every run is done by epoch 300 to 700 and the long-budget campaign is over; stopping on Pearson says the curves have not turned at 19,000 (section 1.5). The manuscript reports Pearson and the objective is the pinball loss, and the metric-aligned round shows the naive resolution, training on Pearson, is unstable at batch 32 and no better than the incumbent at batch 64 (section 2.3). The decision is about what the paper claims, not about what to run.

The incumbent's embedding against ProtT5.. The grid measured content against random at width 1,024 and did not include `calm` (section 2.1). A two-arm replicate set at the incumbent configuration, `calm` against `prot_T5_all`, three seeds each at 1,000 epochs where the replicate spread is 0.0099, resolves a gap of about 0.02 and is the cheapest contrast left on the table: six runs of under a day each.

The per-gene readout at a resolving replicate count.. +0.027 at two pairs against a resolution of 0.06 (section 2.2). Detecting 0.03 at the 4,000-epoch spread of 0.0171 needs about six pairs; at 1,000 epochs and 0.0099 it needs three. Whether the arm's early peak and give-back survive a shorter budget is what a 1,000-epoch replicate would also show.

The Pearson round.. The three cabbi cards holding collapsed batch-32 runs can be freed now or left to the wall on 2026-09-11; the packed tasks cannot be partially canceled without killing their live sibling, and the live `pearson_mse` seed sits in one of them. The batch-64 runs are the only informative survivors and are solo.

Packing.. Four of four packed five-day tasks died of host memory (section 2.4). Until the growth is identified, a five-day arm is either solo on its card or budgeted at 48 GB per run rather than 30, and either choice halves the slots the launch arithmetic assumed.

The per-gene mean on squared error.. No live run beats `nmse` = 1 at its Pearson peak (section 1.1). The retrospective's one-parameter rescale argument has not been applied to any run of the campaign; it costs nothing and would say whether the campaign's models are, like the sprint's, spread 1.8× wider than their correlation justifies.